Subagent-Driven Development (SDD)

A complete multi-agent workflow for turning ideas into shipped code.

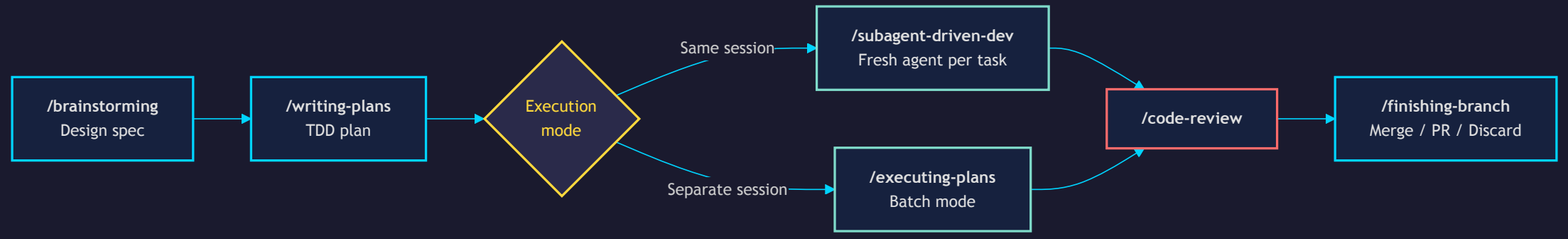
7 skills that chain together: design, plan, execute, review, ship.

The 7 Skills

#	Skill	Purpose
1	`/brainstorming`	Design gate. Explore, question, propose, write spec.
2	`/writing-plans`	Convert spec into bite-sized TDD implementation plan.
3	`/subagent-driven-development`	Execute via fresh subagent per task + two-stage review.
4	`/executing-plans`	Simpler batch execution in a separate session.
5	`/requesting-code-review`	Dispatch code reviewer with clean context.
6	`/finishing-a-development-branch`	Verify tests, merge/PR/keep/discard.
7	`/using-git-worktrees`	Create isolated workspace with safety checks.

`/using-git-worktrees` runs before any implementation step.

The Workflow



Skill 1: `/brainstorming`

Hard gate before any implementation. No code without an approved spec.

1. Explores project context (files, docs, recent commits)
2. Asks clarifying questions (one at a time, multiple choice preferred)
3. Proposes 2-3 approaches with trade-offs
4. Writes spec to `docs/specs/YYYY-MM-DD-<topic>-design.md`
5. Self-reviews for placeholders, contradictions, scope creep
6. User reviews the written spec
7. Transitions to `/writing-plans` (the only exit)

YAGNI ruthlessly. Design for isolation and clarity.

Skill 2: `/writing-plans`

Turn approved specs into construction manuals. Assume the implementer has zero codebase context.

Bite-sized steps (2-5 min each), every step is ONE action:

```
- [ ] Step 1: Write the failing test
      (actual test code here)
- [ ] Step 2: Run it, verify it fails
      Run: pytest tests/test_foo.py -v
      Expected: FAIL with "function not defined"
- [ ] Step 3: Implement minimal code to pass
      (actual implementation code here)
- [ ] Step 4: Run test, verify it passes
- [ ] Step 5: Commit
```

Skill 2: No Placeholders Rule

These are **plan failures**. Never write them:

- "TBD", "TODO", "implement later", "fill in details"
- "Add appropriate error handling"
- "Write tests for the above" (without actual test code)
- "Similar to Task N" (repeat the code)
- Steps that describe what to do without showing how

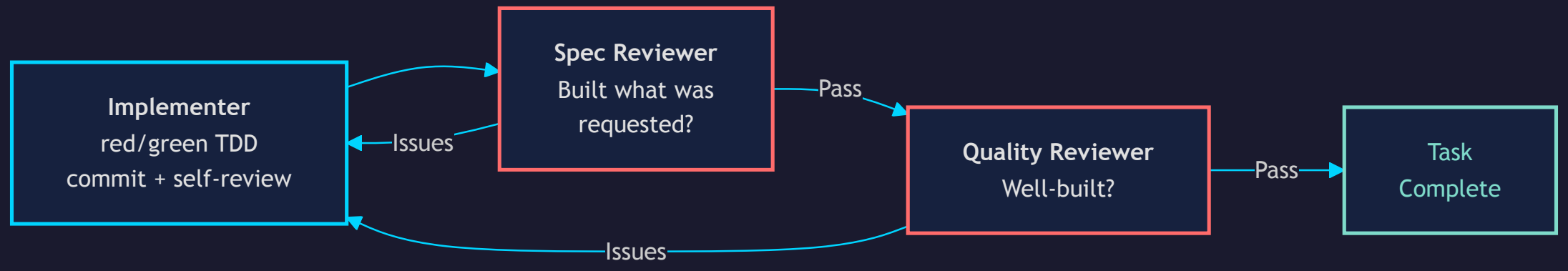
Every step has: **actual code, exact commands, expected output.**

Self-Review Checklist

1. Spec coverage: can you point to a task for every requirement?
2. Placeholder scan: any red flag patterns?
3. Type consistency: do names match across tasks?

Skill 3: `/subagent-driven-development`

Execute the plan in the current session. Fresh subagent per task.
Two-stage review after each.



Skill 3: Model Selection

Use the least powerful model that can handle each role:

Task Type	Signal	Model
Mechanical	1-2 files, clear spec, isolated	Fast/cheap (Sonnet)
Integration	Multi-file, coordination needed	Standard (Sonnet)
Architecture/Review	Design judgment, broad context	Most capable (Opus)

Implementer Status Handling

Status	Action
DONE	Proceed to spec review
DONE_WITH_CONCERNS	Read concerns, address if needed
NEEDS_CONTEXT	Provide missing info, re-dispatch
BLOCKED	More context, better model, or smaller pieces

Skill 4: `/executing-plans`

Simpler alternative. Separate session, batch mode, periodic review checkpoints.

Process

1. Load and critically review the plan
2. Raise concerns before starting
3. For each task: mark in-progress, follow steps exactly, run verifications
4. **Red/green TDD for every task** (tests before implementation, always)
5. When done: `/finishing-a-development-branch`

When to Stop

- Hit a blocker (missing dependency, test fails, unclear instruction)
- Plan has critical gaps
- Verification fails repeatedly

Ask for clarification rather than guessing.

Skill 5: `/requesting-code-review`

When (mandatory)

- After each task in subagent-driven development
- After completing a major feature
- Before merge to main

How

1. Get git SHAs (base and head)
2. Dispatch code-reviewer subagent with: what was built, the plan, the diff

Output

- **Strengths** (be specific)
- **Issues**: Critical (must fix) / Important (should fix) / Minor (nice to have)
- **Assessment**: Ready to merge? Yes / No / With fixes

Skill 6: `/finishing-a-development-branch`

Process

1. **Verify tests pass** (cannot proceed if they fail)
2. **Determine base branch**
3. **Present exactly 4 options:**

Option	What Happens
1. Merge locally	Checkout base, pull, merge, test merged result, delete branch
2. Push + PR	Push with `-u`, `gh pr create`, display URL
3. Keep as-is	Report status, leave worktree
4. Discard	Requires typed confirmation, deletes branch

4. **Cleanup worktree** (for options 1, 2, 4)

Skill 7: `/using-git-worktrees`

Isolated workspace with safety verification.

Directory Selection

1. Check for `.worktrees/` (preferred) or `worktrees/`
2. Check CLAUDE.md for preference
3. Ask user

Safety (Critical)

Verify directory is in `.gitignore` before creating. If not, add it and commit first.

Setup

Auto-detects project type and runs the appropriate install command
(`npm install`, `pip install`, `cargo build`, `go mod download`).
Then runs baseline tests to verify clean state.

Design Principles

1. **Design-first gate**: no code without an approved spec
2. **Two-stage review**: spec compliance (right thing?) then code quality (built well?)
3. **Red/green TDD**: unconditional. Tests before implementation, always.
4. **No placeholders**: every plan step has real code, real commands, expected output
5. **Fresh subagents per task**: prevents context pollution
6. **Bite-sized steps**: 2-5 min granularity, one action per step
7. **Isolation via worktrees**: no branch-switching confusion
8. **YAGNI**: remove unnecessary features ruthlessly

Applying SDD to CW Forge

Using the skills to build a spec-to-MR coding harness.

CW Forge: The Situation

The Plan

`~/.claude/plans/effervescent-chasing-robin.md`

Flask UI + deterministic FSM orchestrator + Claude Code headless agents + parallel coding via git worktrees. **5 milestones**, MVP through polish.

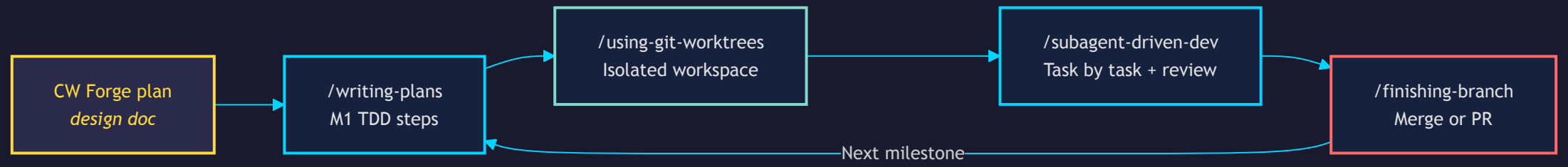
The Gap

The plan is a **design document** (what to build and why).
SDD needs a **construction manual** (exactly what to type, in what order).

The plan already has: architecture diagram, design decisions, SQLite schema, UI layout, directory structure, reuse map, risk mitigations.

This IS the spec. Brainstorming = done. Next: `/writing-plans`.

CW Forge: The SDD Workflow



CW Forge: What Changes

Before (design doc checkbox)

> `state.py: SQLite schema, init_db(), CRUD for run/task/event`

After (/writing-plans output)

```
- [ ] Step 1: Write failing test for init_db()
      def test_init_db_creates_all_tables():
          db = init_db(":memory:")
          tables = db.execute("SELECT name FROM sqlite_master").fetchall()
          assert {"run","task","event","chat"} == {t[0] for t in tables}

- [ ] Step 2: Run test, verify it fails
      Run: pytest tests/test_state.py::test_init_db -v
      Expected: FAIL with "cannot import name 'init_db'"

- [ ] Step 3: Implement init_db()
      (actual code with full CREATE TABLE statements)
```

One checkbox becomes 15-20 executable TDD steps.

Summary

The design stays the **source of truth**.
The construction manual is **disposable** and regenerated per milestone.

For each milestone:

`/writing-plans` → `/using-git-worktrees` → `/subagent-driven-development` → `/finishing-a-development-branch`

What SDD enforces:

- No code without a spec (brainstorming gate)
- No implementation without a plan (writing-plans gate)
- Red/green TDD on every task (unconditional)
- Two-stage review after every task (spec + quality)
- Fresh context per task (no pollution)
- Clean finish (tests, merge/PR, worktree cleanup)